HIGHWAY ACCIDENT ANALYSIS IN INDIA

Official road safety statistics | Calendar years 2018-2022

End-to-end data visualization project

ADS Foundation Workshop
Prepared with SQL, Python and a Power BI dashboard design

Data source: Ministry of Road Transport and Highways (MoRTH), Government of India

1. Abstract

This project analyzes official Indian road accident statistics to understand the National Highway (NH, including Expressways) accident trend, compare reported NH counts across the states ranked highest in 2022, and place highway outcomes in the context of all-road totals. The source tables provide annual aggregates rather than individual crash records. Python is used to reshape and validate the published data, SQL provides a relational model and reusable queries, and the Power BI package defines a dashboard model, measures and layout.

The published series shows 151,997 NH accidents in 2022, an increase of 18.0% over 2021 and 7.9% over 2018. Tamil Nadu recorded the highest NH accident count among the 2022 top-ten states (18,972), followed by Kerala (17,627). NH/Expressways represented 32.94% of all reported accidents and 36.22% of fatalities in 2022. These are counts and shares, not exposure-adjusted risk estimates.

2. Problem statement and objectives

Road safety decisions need clear, comparable evidence. Published accident tables are spread across annual reports and annexures, and raw counts can be misread without matching road categories and clearly stating coverage. This project organizes those statistics into analysis-ready tables and a dashboard plan.

- Track the national NH accident trend across 2018-2022.
- Identify the states with the highest NH accident counts in the MoRTH 2022 top-ten table.
- Compare 2022 NH accident and fatality shares with all-road totals.
- Demonstrate SQL aggregation and joins, Python data cleaning and EDA, and Power BI model and measure design.

3. Dataset description

The project uses the published MoRTH Road Accidents in India 2022 report and two PIB releases issued by MoRTH. Calendar-year data are used. State/UT all-road accidents, fatalities and injured persons cover the published 2018-2022 annexures. NH accident totals cover India for 2018-2022. State-level NH accident counts cover only the ten states ranked highest in 2022, with their annual counts from 2018-2022. NH fatalities are included for 2022, where the source provides a matched national figure.

Table	Grain and coverage	Use
State all-road	State/UT-year; 2018-2022	Broad state trends in accidents, fatalities and injuries
India NH annual	India-year; 2018-2022	National NH accident trend; 2022 NH fatalities
NH top-10 state	State-year; 10 states x 5 years	NH state ranking and selected-state trends
India all-road annual	India-year; 2018-2022	National road-category comparisons

The source does not provide individual accident date/time, district or route coordinates, vehicle type, cause, or matched NH injury counts in the tables used here. The top-ten state NH table must not be treated as a full state dataset or summed to create the national total.

4. Methodology and data preparation

The workflow starts with source-faithful raw CSV tables. The state all-road annexures are transcribed into a single wide source table and transformed into one state-year row per published state/UT. Separate tables are retained for national NH totals, all-road national totals, and the NH top-ten states to prevent accidental mixing of road categories or aggregation levels.

Cleaning and quality controls

- Trim state labels and convert year and count fields to numeric types.
- Preserve source blanks as null values. Null means unavailable, not zero; no imputation is performed.
- Check the state-year primary key for duplicates and retain source lineage on each table.
- Use source footnotes for category changes: Daman & Diu is included with Dadra & Nagar Haveli in 2020-2022; Jammu & Kashmir includes Ladakh for 2018-2020, then Ladakh is reported separately.
- Validate key national totals against official published totals before analysis.

Analysis techniques

Python uses Pandas to load, reshape, clean, aggregate and export CSV tables. NumPy supports safe calculation of ratios with missing values. Descriptive analysis calculates year-over-year NH accident change, selected-state totals and shares, 2022 state rankings, and fatalities per 100 accidents where matched counts exist. SQL tables enforce primary keys and sensible nonnegative count checks. SQL queries use GROUP BY, JOIN, CASE and window functions for ranking, KPI calculations and year-over-year comparison.

5. SQL analysis

The SQL model includes separate fact tables for state all-road year, top-ten state NH year, India NH year, and India all-road year. This star-friendly separation avoids summing state aggregates into national totals when the NH state table is intentionally limited. The query pack includes the national annual NH trend with LAG, 2022 state ranking and national share, the matched 2022 NH/all-road comparison, and a 2018-to-2022 state comparison.

6. Python analysis and key findings

Indicator	Finding
NH accidents, 2022	151,997
Change from 2021	+18.0% (128,825 to 151,997)
Change from 2018	+7.9% (140,843 to 151,997)
Top 10 state NH accidents, 2022	116,099, or 76.4% of the national NH total
NH share of all-road accidents, 2022	32.94% (151,997 / 461,312)
NH share of all-road fatalities, 2022	36.22% (61,038 / 168,491)

In 2022, Tamil Nadu had the highest NH accident count among the ten states shown (18,972); Kerala was second (17,627). National NH accidents fell to 116,496 in 2020, then rose in 2021 and 2022. The 2022 national count exceeded the pre-pandemic 2018 count. The comparison indicates a higher share of fatalities than accidents on NHs, but does not establish why this difference exists.

7. Power BI dashboard design

The supplied Power BI build guide and DAX file are implementation materials for Power BI Desktop. The recommended single report page includes KPI cards for 2022 NH accidents and fatalities and the NH shares of all-road accidents and fatalities; a line chart for the national NH trend; a ranked horizontal bar chart for the 2022 top-ten states; and a state/year slicer for selected-state comparisons. The footer should display source and scope notes.

Model relationships connect a year dimension to each fact table and a state dimension to the two state-level tables. Measures for national totals use India-level facts. State comparison visuals must use the top-ten NH table and be labelled accordingly. The project package supplies a static visual preview as a layout reference; it is not an interactive Power BI report file.

8. Findings and interpretation

- Reported NH accidents increased sharply in 2022: 18.0% above 2021 and 7.9% above 2018.
- The top ten states in the MoRTH 2022 NH ranking contributed 76.4% of national NH accident counts in 2022.
- NHs/Expressways accounted for 32.94% of all-road accidents but 36.22% of fatalities in 2022.
- These data support descriptive comparisons only. Road length, traffic volume, vehicle-kilometres and individual crash factors are not available in the supplied tables.

9. Conclusion

The project turns official annual road safety statistics into reproducible analysis tables and a clear dashboard plan. Its main message is that national NH accident counts rebounded in 2022 and that NH fatalities formed a larger share of all-road fatalities than NH accidents formed of all-road accidents. Counts should be presented with their year, road category, and coverage label so that the audience does not confuse recorded volume with underlying risk.

10. Future scope

- Obtain district-, corridor- or route-level e-DAR records to identify local accident concentrations.
- Add traffic volume, road length or vehicle-kilometres to estimate exposure-adjusted rates.
- Add time, cause, vehicle, road condition and injury-severity fields only when a reliable source provides them.
- Refresh the model as new MoRTH annual reports become available, and document any definition or boundary changes.

Sources

Ministry of Road Transport and Highways, Road Accidents in India 2022, including Table 5.5 (top ten states on National Highways, 2018-2022): https://morth.gov.in/sites/default/files/RA_2022_30_Oct.pdf

PIB / MoRTH, Road Accidents and Safety Measures, 7 August 2024, national/state all-road series and 2022 NH comparison: https://www.pib.gov.in/PressReleseDetailm.aspx?PRID=2042509&lang=2&reg=48

PIB / MoRTH, Prevention of Road Accidents in the Country, 26 July 2023, NH annual accident counts 2018-2021: https://www.pib.gov.in/PressReleasePage.aspx?PRID=1942889&lang=2&reg=48